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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 524 — Past Questions & Materials  
**by Renance**

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Course: MAR 524  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 5  
Levels: 500

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## **Paper 1 of 5 — MAR 524 MARINE STEEM AND GAS TURBINE SECOND SEMESTER 2023/2024**

Level 500 • Second Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN**  
**DEPARTMENT OF MARINE ENGINEERING**  
**MARINE STEAM AND GAS TURBINE (MAR 524) ASSESSMENT**  
**(BATCH A)**

Time: 2HRS.

**PROBLEM SET**

DONGFANG INC simple Rankine power plant (GET240) efficiency can no longer satisfy the needs of OTAMARA community in BAYELSA STATE due to increasing consumption of power by end users. A thermal plant with higher capacity is required to serve the community needs. Thus, this prompted DONGFANG INC. to design/manufacture a regenerative thermal plant (REG401) that would operate in REGENERATIVE CYCLE.

As an EXPERT in marine steam turbine operations:

- I. Suggest two advantages of REG 401 over GET 240. (1 mark)
- II. Explain the working principles of the regenerative cycle with the aid of a working diagram and describe the likely drawbacks associated with the cycle (3 marks)
- III. Using the working diagram and T-S DIAGRAM below in Fig. 1:

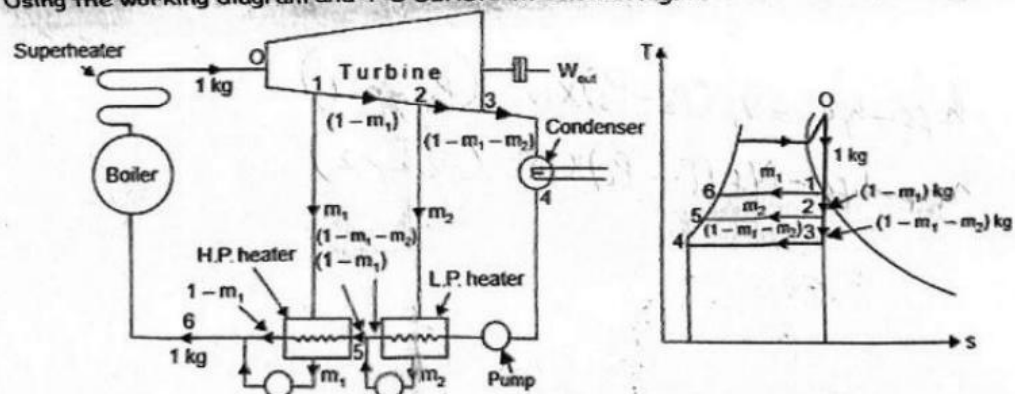


Fig. 1: Regenerative cycle

- a. Provide the governing energy equations for the H.P & L.P turbine (4 marks)
- b. Assuming REG401 operate as a SINGLE HEATER regenerative cycle where steam enters the turbine at 35 bar and 400 degrees Celsius with exit pressure of 0.10bar and the feedwater heater is a direct contact type which operate at 5 bar, deduce the efficiency of the thermal plant and increase in mean temperature of heat addition and show by disparity the percentage rise due

## **Paper 2 of 5 — MAR 524 MARINE STEEM AND GAS TURBINE SECOND SEMESTER 2022/2023**

Level 500 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MARINE ENGINEERING

MARINE STEAM AND GAS TURBINE (MAR 524)

SECOND SEMESTER EXAMINATION

Attempt question one (1) and any other four (4) questions

Time: 3 HRS.

Date: 25/09/2023

### QUESTION ONE

- a) A direct mixing type steam turbine is fed with steam having an enthalpy of 3100 KJ/Kg. It moves out of the turbine with an enthalpy of 2100Kj/kg. Feed heating is done at a pressure of 3.2 bar with steam enthalpy of 2500 KJ/kg. The condensate from a condenser with an enthalpy of 125KJ/kg enters into the feedwater. The quantity of bled steam is 11200 Kg/h. Produce a diagram for this process and deduce the power developed by the turbine (8 marks)
- b) In a regenerative steam cycle employing two close feed heaters, the steam is supplied to the turbine at 40 bar and 500°C and is exhausted to the condenser at 0.035 bar. The intermediate bleed pressure is obtained such that the saturation temperature intervals are approximately equal giving of 10 and 1.1 bar. Calculate the amount of steam bled at each stage, the work output of the plant per kilogram of boiler steam and the cycle efficiency of the plant assuming ideal processes where required using Fig. 1 (12 marks)

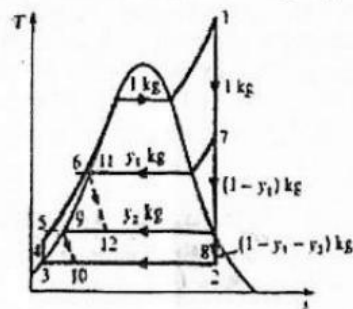


Fig. 1: Two feed heaters Regenerative cycle

### QUESTION TWO

DONGFANG INC simple Rankine power plant (GET240) efficiency can no longer satisfy the needs of OTAMARA community in BAYELSA STATE due to increasing consumption of power

by end users. A thermal plant with higher capacity is required to serve the community needs. Thus, this prompted DONGFANG INC. to design/manufacture a regenerative thermal plant (REG401) that would operate in RANKINE CYCLE.

As an EXPERT in marine steam turbine operations:

- a) Explain the working principle of a regenerative cycle (1.5 mark)

Using the working diagram and T-S DIAGRAM below in Fig. 2:

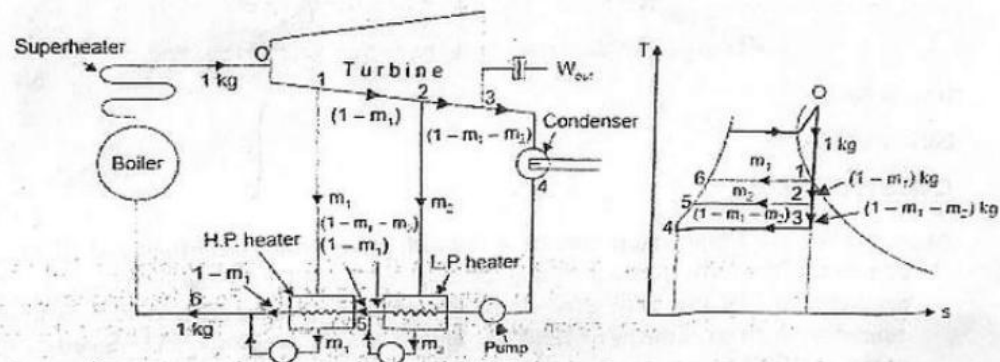


Fig. 2: Regenerative cycle

- Neglecting pump work, provide the thermal efficiency equation for the regenerative cycle (3 marks)
- Assuming REG401 operate as a SINGLE HEATER regenerative cycle where steam enters the turbine at 30 bar and 400 degrees Celsius with exit pressure of 0.10 bar and the feedwater heater is a direct contact type which operate at 5 bar, deduce the efficiency of the thermal plant and increase in mean temperature of heat addition and show by disparity the percentage rise due to regeneration (where the enthalpy of the steam at the point of entering FWH is 2737.99 KJ/KG) (8 marks)

### QUESTION THREE

In a steam turbine, steam leaves the boiler plant at 20 bar, 300°C and expanded to 0.035 bar before entering the condenser. Compare the following cycles in terms of efficiencies and specific steam consumption for:

- A superheated Rankine cycles (2.5 marks)
- A reheat cycle with steam reheated to 300°C at the pressure when it becomes saturated vapour. (5 marks)

$$W_T = m c_p (T_2' - T_3)$$

- iii. With the aid of a well labelled diagram, explain the working operation of a superheated Rankine cycle and provide the governing equations for boiler, turbine, condenser and feed pump work assuming no pressure drop occurs (2.5 marks)
- iv. Applying the governing equation obtain in "iii", rework superheated Rankine cycle efficiency when steam leaves the boiler at 20bar and 300°C and expanded to 0.035 bar as stated in "i" (Take  $\eta_T = 0.9$ ,  $\eta_{wp} = 0.85$ ) (2.5 marks)

#### QUESTION FOUR

In a combine gas steam plant cycle with power output of 450MW, the gas turbine compression ratio is 14. Air enters compressor at 300K and turbine inlet at 1400K. The combustion gases leaving the gas turbine are used to heat the steam at 8 Mpa to 673K in the heat exchanger (HE). The combustion gases leave the HE at 480K. The condenser pressure is 20Kpa. Assume isentropic efficiencies of compressor and turbine to be 81.5 % & 85.5%. Assuming gamma ( $\gamma$ ) = 1.4, deduce:

- a) Ratio of Mass flow rate of steam to gas (6 marks)
- b) Heat input into the combustion chamber  $\times 10^4$  (4 marks)
- c) Thermal efficiency of the combined plant (2.5 marks)

#### QUESTION FIVE

- a) A Carnot engine whose low temperature reservoir is at 7-degree Celsius has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees the temperature of the source be increased? (4 marks)
- b) Air enters the compressor of an open cycle constant pressure gas turbine at a pressure of 1 bar and temperature of 20°C. The pressure of the air after compression is 4 bar. The isentropic efficiencies of compressor and turbine are 80% and 86% respectively. The air-fuel ratio used is 89:1. If the calorific value & power developed of fuel is 41800 Kj/kg and 252.84 KW/Kg of air. Deduce:
- i. Mass flow rate in kg/sec  $m_g c_{p_g} (T_{g1} - T_{g2}) = m_s (h_{f8} - h_{f3})$  (4.25 marks)
- ii. Thermal efficiency of the cycle (4.25 marks)

$$(Take \gamma = 1.4, C_p = 1.01 \frac{KJ}{kg} K \text{ for air and gas})$$

#### QUESTION SIX

- a) A marine power plant operating on an ideal Brayton cycle has a pressure ratio of 8. The gas temperature is 302K at compressor inlet and 1350K at turbine inlet. Assuming compressor and turbine efficiencies are 80% and 86% respectively. Deduce the back work ratio and thermal efficiency assuming  $\gamma = 1.4$  for air and gas (8 marks)
- b) Explain the concept of Joule Brayton cycle and deduce the condition for air standard assumption efficiency for an open simple gas turbine using an example (4.5 marks)

## QUESTION SEVEN

- a) In a Rankine turbine, steam at inlet to turbine is saturated at a pressure of 45 bar and the exhaust pressure is 0.2 bar. Assuming flow rate of 9.5 kg/sec, determine:
- Pump work
  - Turbine work
  - Rankine efficiency
  - Condenser work and
  - Dryness at the end of expansion (1.5 mark each)
- b) An engineer proposed a system of two Carnot cycles, a heat engine and a heat pump to transfer heat from 400 K to 500 K. The heat engine has high and low temperature of 400 and 300 K respectively. Is this process possible? (5 marks)

$$\frac{T_2}{337} = 0.815$$

$$T_2 = 300$$

$$337 = 0.815 T_2' - (0.815 \times 300)$$

$$337 +$$

Course Lecturer:

CLEMENT A.I

MNIMENA

## Paper 3 of 5 — MAR 524 SHIP POWER PLANT 11 SECOND SEMESTER 2023/2024

Level 500 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MARINE ENGINEERING

SHIP POWER PLANT II (MAR 524)

ATTEMPT THREE (3) QUESTIONS. QUESTION NO.1 IS COMPULSORY

TIME:90 MINS

1. Carnival cruise line was sailing with full crews onboard in the Atlantic Ocean. Distilled water is required for various ship services such as cooling, washing/cleaning, cooking and running of other important machinery. Hence:

- Propose a system that can satisfy these needs and state by prediction the required amount of fluid needed in liters daily for Carnival cruise line (1 mark)
- Provide a working diagram showing the various parts of the proposed systems and explain how it achieves its working operations. ( marks)
- What are the three methods that can be used in producing desalinating water from your proposed system? (15 mark)
- Describe the arrangement of the proposed system with respect to two compartments (0.5 mark)
- How can scale formation be minimized in the proposed system (1 mark)
- Explain the role of air-purge valve and ejector in your proposed system (1 mark)

2. a) With respect to cooling systems for ship power plants, explain direct and indirect cooling systems and discuss their merits and demerits on ship machinery \_\_ (6.5 marks)

b) Develop a seawater cooling diagram for a slow speed Engine (3.5 marks)

3; a) State the relevance of steam system in ship with respect to fuel treatment, fuel purification and domestic applications. (6 marks)

b) State the difference between main boiler and auxiliary boiler (4 marks)

4. a) Explain the concept of Unattended Machinery space (UMS) in ship and discuss the need for total integrated systems (2 mark)

b) State the significance of total integrated systems on the following ship systems:

r. Ship main engine control (2 marks)

Ir. Bridge control (2 marks)

TIT. Cargo and ballast monitoring & control (2 marks)

c) Discuss the significance of automatic and control systems in ship with respect to watchkeeping 5 (2 marks)

COURSE LECTURER:

CLEMENT I.A,

## **Paper 4 of 5 — MAR 524 MARINE SYSTEM AND GAS TURBINE SECOND SEMESTER 2022/2023**

Level 500 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MARINE ENGINEERING

MARINE STEAM AND GAS TURBINE (MAR 524)

SECOND SEMESTER EXAMINATION

Attempt question one (1) and any other four (4) questions

Time: 3 HRS.

Date: 25/09/2023

### QUESTION ONE

- a) A direct mixing type steam turbine is fed with steam having an enthalpy of 3100 KJ/Kg. It moves out of the turbine with an enthalpy of 2100Kj/kg. Feed heating is done at a pressure of 3.2 bar with steam enthalpy of 2500 KJ/kg. The condensate from a condenser with an enthalpy of 125KJ/kg enters into the feedwater. The quantity of bled steam is 11200 Kg/h. Produce a diagram for this process and deduce the power developed by the turbine (8 marks)
- b) In a regenerative steam cycle employing two close feed heaters, the steam is supplied to the turbine at 40 bar and 500°C and is exhausted to the condenser at 0.035 bar. The intermediate bleed pressure is obtained such that the saturation temperature intervals are approximately equal giving of 10 and 1.1 bar. Calculate the amount of steam bled at each stage, the work output of the plant per kilogram of boiler steam and the cycle efficiency of the plant assuming ideal processes where required using Fig. 1 (12 marks)

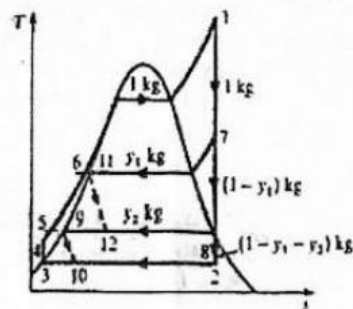


Fig. 1: Two feed heaters Regenerative cycle

### QUESTION TWO

DONGFANG INC simple Rankine power plant (GET240) efficiency can no longer satisfy the needs of OTAMARA community in BAYELSA STATE due to increasing consumption of power

**DEPARTMENT OF MARINE ENGINEERING**  
**RUNNING AND MAINTENANCE OF SHIP POWER PLANT (MAR 521)**  
**END OF SEMESTER TEST**

**TIME: 90 MINS.**

**QUESTION ONE**

- engine, fuel oil, lubrication*
- ✓ a. What constitutes "ship machinery spaces"? (3.5 marks)
  - ✓ b. Discuss the role of main engines and auxiliaries onboard a ship (2.5 marks)
  - ✓ c. What do you understand by the term "Machinery space tools"? (0.5 mark)
  - ✓ d. Discuss two relevance of machinery spaces tools (1 mark)
  - ✓ e. Enumerate five basic tools for ship machinery's repair (2.5 marks)

**QUESTION TWO**

- ✗ a. Describe the investigation and remedial action required if a ship main engine turns on air but fails to fire on fuel? (2 marks)
- ✓ b. With respect to machinery space knowledge, discuss the core functions of the engineering department onboard a ship (1 mark)
- ✓ c. How can we minimize maintenance tasks on ship machinery spaces? (1 marks)
- ✓ d. What ensures high-functioning, clean and safe operations in ship's machinery spaces? (1 mark)
- ✓ e. Discuss three possible ultimate yardsticks that can be used to measure the competence of a watchkeeping procedure that can improve the performance of a marine engineer (3 marks)
- ✓ f. Having been taught "Running and maintenance of ship power plants" discuss the relevance of this knowledge on any Marine Systems of your choice (2 marks)

*main engine for propulsion*

*power generation*

*shear 2 ships*

**CLEMENT A.I**  
**MNIMENA**

## Paper 5 of 5 — MAR 524 MARINE SYSTEM AND GAS TURBINE SECOND SEMESTER 2022/2023

Level 500 • Second Semester • PastQuestion

FEVERAL UNNIVCNOLLY OF PEFROLEUM RESOURCES, EFFURUN  
COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester

SESSION: 2022/2023

DATE: il Mebied

COURSE CODE: EEE 525

COURSE TITLE: Mechatronics II

Leas ALLOWED: 2Hr 15mins

21 (a) List two Merits and demerits of potentiometer. ; 1 (4 marks)

| -(b) A linear resistance potentiometer is 200 mm long and is uniformly wound with a wire of total resistance 0.005-MQ. Under normal conditions, the slider is at the centre of the potentiometer. Determine the linear displacement when the resistance of the potentiometer measured by wheatstone bridge is 0.0000015 GQ. If it is possible to measure a minimum value of 5 Q resistance with the above arrangement, determine the resolution of the potentiometer in metres. (6 marks)

,--(c) An LVDT is employed for measuring the deflection of a bellow. The sensitivity of the LVDT IS 600 V/mm. The bellow is deflected by 0.155 mm by a pressure of  $1.2 \times 10$  N/m?. Determine the sensitivity of the LVDT in volts per N/m? and the pressure when the output voltage is 4.5 V (4 marks)

S/d) A pump supplies oil at 126 LPM to a 6 cm diameter double-acting hydraulic cylinder. If ' the load is 0.00001 GN (extending and retracting) and the rod diameter is 3 cm, find

- (i) Hydraulic pressure during the extending stroke. (2 marks)
- (ii) Piston velocity during the extending stroke. (2 marks)
- (iii) Cylinder power in Watts during the extending Stroke. (2 marks)
- (iv) Hydraulic pressure during the retracting stroke. (2 marks)
- (v) Piston velocity during the retracting stroke. (2 marks)

Q2. a(i) Give four advantages and disadvantages of a Linear Variable Differential Transformer. (2 marks)

(ii) State and briefly explain five requirements every transducer must meet. (5marks)

A strain gauge is having a resistance of 100 Q and a gauge factor of 2 is connectec in series with a ballast resistance of 100 across 12 V supply. Calculate th difference between the output voltage when:

- (a) No stress is applied (3marks)
- (b) A stress of 104 MN/m? is applied

Note: The modulus of elasticity is 200 GN/m? (2marks)